

ABOUT ME

Software engineer and researcher working on Quantum Computing. Experienced in developing full-stack applications, optimizing machine learning models and quantum computing algorithms. Background in containerization, and automation for AI-driven applications in Python and Rust. Passionate about bridging research and software development to create efficient, secure, and innovative solutions.

EDUCATION

Institute of Science Tokyo (Former Tokyo Institute of Technology) Ph.D. in Information and Communication Engineering (<i>Super Smart Society Incentive</i>)	<i>Sept. 2023-Present</i>
The High Energy Accelerator Research Organization (KEK) M.Sc. in Experimental High Energy Physics (<i>Monbukagakusho Honors</i>)	<i>Oct. 2020-Sept. 2023</i>
University of California, San Diego B.Sc. in Physics (<i>Provost Honors</i>)	<i>June 2016-Mar. 2020</i>

TECHNICAL SKILLS

Programming: Python, C++, Rust, Typescript, Kotlin, Julia

Packages or Libraries: PyTorch, JAX, PennyLane, Qiskit, Manifolds.jl/Manopt.jl, Scikit-learn, Flask, React, GEANT4

Dev Tools: Docker, Grafana, MongoDB, Portainer, PostgreSQL

Languages: English (Professional), Japanese (Business Intermediate), Mandarin (Native)

EXPERIENCE

Research Associate RIKEN Advanced Intelligence Project Developed methods for defending and attacking large language and vision-language models. Conducted theoretical research on AI security and privacy.	<i>Apr. 2024-Present</i>
Visiting Scholar HKUST-Guangzhou Developing package with rigorous mathematical foundation and scientific computing tools in quantum computing, and provide issues for problem reduction in complexity theory.	<i>Jan. 2026-Mar. 2026</i>
Software Engineer Intern ExaWizards Inc. Developed a BtoB web application for enterprise HR skill training using React and Kotlin. Implemented database migration and data flow design to enhance system efficiency and user experience.	<i>Aug. 2025-Dec. 2025</i>
Junior Physicist Inner Tracking Detector Group (ATLAS Exp., CERN) Analyzed Higgs self-coupling physics ($HH \rightarrow b\bar{b}\tau\tau$) using machine learning techniques. Collaborated with CERN on detector testing and data interpretation.	<i>Apr. 2021-Sept. 2023</i>
Lab Assistant Popmintchev Lab, UCSD Designed optical systems for high harmonic generation. Simulated intensity profiles of laser beams using Matlab for ionized gases (Ar^+ , He^+ , Ne^+).	<i>July 2018-Mar. 2020</i>

PROJECTS

Problem-reduction: Extensible framework for Computational Complexity Professional scientific software, exploring automation of Agentic Engineering in Computational Complexity. Build up a pipeline for human-agent working system to explore the computational complexity. lib <i>Tools: Opus4.6, Claude Code, Rust</i>	<i>Jan. 2026-April 2026</i>
pdfT: Trainable Sparse Bases for Image Compression and Inpainting Authored pdfT , a JAX library learning trainable sparse bases that replace the DFT/DCT, optimized on the unitary manifold so the transform stays exactly invertible (QFT, entangled-QFT, TEBD, MERA, DCT-IV bases). Trained bases store Quick Draw images in roughly 20% fewer bytes than JPEG's 8×8 block cosine transform at equal quality (arXiv:2608.00053); with frozen Hadamards the same circuits inpaint DIV2K images 2 dB above the DFT using 288 phases (arXiv:2609.17298). Supersedes the Julia original ParametricDFT.jl, ported to feature parity against committed goldens. lib bench <i>Tools: Python, JAX, Julia, Manopt.jl, Manifolds.jl, Docker</i>	<i>Dec. 2025-Present</i>
ManifoldsGPU.jl: GPU-Accelerated Riemannian Optimization Software contribution to the JuliaManifolds organization bringing CUDA acceleration to manifold optimization: batched GPU kernels for manifold operations and retractions, giving roughly $20\times$ speedup for <code>exp!</code> on <code>PowerManifold(Stiefel(32,16), 2048)</code> and $15\times$ for batched-SVD polar retraction. lib <i>Tools: Julia, CUDA.jl, Manifolds.jl, Manopt.jl</i>	<i>Jan. 2026-Present</i>

LogosQ: Quantum Computing Library in Rust

July. 2025-Present

Developed a quantum computing library in Rust, implementing quantum gates, circuits, and simulation algorithms from scratch. Optimized the gate with direct gate manipulation and Tensor Contractions, lead to publication and submission to IEEE Transaction on Quantum Engineering. lib doc.

Tools: Cargo, Rust, Docker, Typescript

DIA: Human Resource Technology for Enterprises Skills Training

August. 2025- Dec. 2025

A BtoB product that leverages AI to provide personalized training and development programs for employees. Meanwhile, it offers analytics and insights to organizations for workforce optimization and automatic skill gap identification, and generate the tickets distributed to Hiring website like Indeed, Glassdoor, and LinkedIn.

Tools: React, Javalin, PostgreSQL, Docker, Kotlin, TypeScript, Postman

Data Extraction and Model Alignment

Nov. 2024-Apr. 2025

Developed scalable data extraction attack algorithms for large language models, improving attack methodologies for membership inference and model alignment. Built end-to-end data pipelines, integrating reinforcement learning to optimize model robustness. Designed monitoring and visualization dashboards with Grafana.

Tools: Grafana, PyTorch, Llama2/3, GPT-2 XL, Docker

Graph Encoding System with Quantum Computing

Sept. 2023-Nov. 2024

Developed a variational quantum circuit for tensor-based graph encoding. Deployed API interfaces for quantum computation workflows. Published research findings on ArXiv and accepted by IEEE conference.

Tools: PennyLane, Qiskit, Python, C++

Data Analysis for Higgs Boson Research

Jan. 2022-Sept. 2023

Engineered a machine learning pipeline for high-dimensional image data from the ATLAS pixel detector, with custom CNNs and GNNs for feature extraction that improved detection by 15 ~ 20%, plus automated preprocessing for large-scale datasets.

Tools: Python, TensorFlow, PyTorch, mySQL

PUBLICATIONS

Quantum-Inspired Trainable and Parameter-Efficient Tensor Networks for Image Inpainting

Preprint, arXiv:2609.17298, 2026. Submitted to ICASSP 2027.

URL: <https://arxiv.org/abs/2609.17298>

Fast Trainable Multilinear Bases for Image Compression

Preprint, arXiv:2608.00053, 2026.

URL: <https://arxiv.org/abs/2608.00053>

LogosQ: A High-Performance and Type-Safe Quantum Computing Library in Rust

IEEE Transactions on Quantum Engineering (Under Review), Techrxiv., 2025.

Tensor-Based Binary Graph Encoding for Variational Quantum Classifiers

2025 IEEE Quantum Computing and Network Communication, Nara, Japan, arXiv:2501.14185, 2025.

URL: <https://arxiv.org/abs/2501.14185>

Search for Nearly Mass-Degenerate Higgsinos Using Low-Momentum Mildly Displaced Tracks in pp Collisions at $\sqrt{s} = 13$ TeV with the ATLAS Detector

Physical Review Letters, Vol. 132, No. 22, American Physical Society (APS), 2024.

DOI: 10.1103/PhysRevLett.132.221801

Overnight Stock Movement Prediction with Contextualized Embedding Incorporating News and Stock

2022 IEEE 11th Global Conference on Consumer Electronics (GCCE), Pages 476-478, 2022.

DOI: 10.1109/GCCE56475.2022.10014070

AWARDS AND HONORS

Science Tokyo Spring Scholarship — next-generation front-runner development program.

2025.12-Now

TSUBAME Scholarship — for outstanding Ph.D. students at Institute of Science Tokyo.

2025.10-12

Super Smart Society Research Incentive — fundamental technologies for a Super Smart Society.

2024.4-Now

JASSO Monbukagakusho Honors Scholarship — government award for graduate study at KEK.

2021-2022

Provost Honors — UCSD Provost's Honor List for undergraduate academic performance.

2016-2020

ADDITIONAL ACTIVITIES

- Contribute to multiple open source projects on GitHub, including PennyLane v0.40.0 and JuliaManifolds.
- Board Game Arena, Catan Global Top 50. + Scythe, Brass Birmingham.
- PADI Advanced Open Water Diver (Licensed 2018).
- GO, Amatuer 2 Dan.